

ERTIS, Kazakhstan

WMO: 298070

Lat: 53.3375N

Lon: 75.4619E

Elev: 308

StdP: 14.53

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.3	-24.7	-35.8	0.7	-30.1	-30.5	1.0	-24.8	27.2	14.0	22.4	12.3	2.8	30	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	89.5	64.9	86.3	64.2	83.1	63.4	67.9	82.2	66.5	80.8	65.1	79.1	6.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.2	87.7	73.0	61.2	81.7	72.3	59.4	76.4	71.6	32.5	82.2	31.3	80.6	30.2	78.8	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.7	20.4	17.2	DB	-34.4	95.1	6.8	3.7	-39.3	97.8	-43.2	99.9	-47.0	102.0	-51.9	104.6	(4)
(5)			WB	-34.5	71.8	6.6	1.6	-39.3	73.0	-43.1	73.9	-46.8	74.8	-51.6	76.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.3	-0.4	4.2	19.4	42.6	56.4	66.8	69.1	65.2	53.9	40.1	21.5	6.9		
	DBStd	27.00	14.39	12.99	11.96	10.69	9.49	7.59	6.46	7.16	8.50	9.09	13.92	13.94		
	HDD50	6681	1561	1283	950	268	46	2	0	1	53	326	854	1337		
	HDD65	10525	2026	1703	1415	672	294	70	33	88	347	771	1304	1802		
	CDD50	2051	0	0	0	46	246	505	593	471	170	20	0	0		
	CDD65	421	0	0	0	1	29	123	162	93	13	0	0	0		
	CDH74	4344	0	0	0	22	414	1236	1520	965	184	3	0	0		
	CDH80	1426	0	0	0	3	110	411	526	324	52	0	0	0		
	WSAvg	6.2	6.0	6.6	7.0	7.2	7.1	5.7	4.7	5.0	5.7	5.8	6.6	6.9		
	PrecAvg	11.5	0.6	0.4	0.4	0.6	1.0	1.5	2.2	1.6	0.8	1.0	0.8	0.6		
	PrecMax	18.3	1.6	1.1	1.3	1.5	2.9	3.9	8.1	3.1	2.6	2.4	2.4	1.4		
(15) Precipitation	PrecMin	5.8	0.0	0.0	0.0	0.0	0.0	0.2	0.3	0.4	0.1	0.3	0.0	0.0		
	PrecStd	2.8	0.3	0.3	0.3	0.4	0.7	0.9	1.7	0.8	0.5	0.6	0.4	0.3		
	DB	34.4	34.6	49.0	77.3	88.1	92.2	94.1	93.0	85.9	70.8	49.4	34.7			
	MCWB	32.2	32.0	42.7	58.8	62.8	65.6	66.2	65.2	62.2	54.2	44.1	32.3			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	29.6	31.7	41.7	70.5	83.1	88.3	89.3	87.6	78.9	64.4	44.4	32.7			
	2% MCWB	27.6	29.5	37.4	55.3	60.8	64.8	64.9	64.5	59.9	51.4	40.4	30.8			
	5% DB	25.3	27.5	37.5	65.5	78.8	84.8	85.8	83.3	73.7	58.8	40.6	29.8			
	5% MCWB	23.4	25.3	34.2	52.8	59.4	63.8	64.7	63.7	57.6	48.4	36.9	27.8			
	10% DB	21.3	23.4	34.9	60.1	74.2	81.3	82.2	79.5	68.4	53.8	37.0	26.3			
	10% MCWB	19.8	21.4	32.0	49.8	57.4	62.5	64.2	62.5	55.1	45.9	33.9	24.5			
	WB	32.1	32.3	42.9	59.8	65.2	69.6	70.4	68.6	64.4	55.0	44.8	32.8			
	MCDB	34.1	34.3	48.3	74.3	83.4	86.5	84.0	84.0	80.8	68.4	48.3	34.2			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	27.6	29.4	37.6	56.2	62.3	67.1	68.4	66.7	61.0	51.8	40.6	30.9			
	2% MCDB	29.5	31.5	41.3	68.7	79.5	81.8	81.9	82.7	76.9	62.9	44.4	32.6			
	5% WB	23.4	25.3	34.4	53.3	60.2	65.5	66.9	65.1	58.3	49.0	37.0	27.8			
	5% MCDB	25.2	27.4	37.3	64.8	76.3	80.1	80.2	80.1	72.0	57.7	40.3	29.7			
	10% WB	19.8	21.4	32.1	50.3	58.2	63.9	65.5	63.3	56.0	46.2	34.0	24.3			
	10% MCDB	21.4	23.4	34.8	59.6	72.6	78.3	78.5	76.9	67.4	53.4	36.8	26.2			
	MDBR	14.5	15.7	15.8	19.7	22.2	20.4	19.5	20.4	20.0	16.2	12.8	14.3			
	5% DB	15.0	14.1	14.1	27.9	28.6	25.3	24.7	26.3	27.6	25.7	14.2	14.3			
(35) Mean Daily Temperature Range	MCWBR	14.3	13.3	11.6	15.9	13.4	9.9	9.0	10.8	13.4	15.6	11.4	13.6			
	MCDBR	14.8	14.5	14.2	27.0	26.0	21.9	20.8	23.9	26.1	23.5	13.5	14.6			
	MCWBR	14.2	13.7	11.9	16.2	13.4	10.1	9.4	11.1	13.5	15.0	11.2	14.1			
	MCDBR	14.2	13.7	11.9	16.2	13.4	10.1	9.4	11.1	13.5	15.0	11.2	14.1			
(40) Clear-Sky Solar Irradiance	taub	0.295	0.326	0.385	0.410	0.411	0.419	0.441	0.428	0.373	0.353	0.333	0.300			
	taud	2.040	1.993	1.934	2.140	2.198	2.257	2.198	2.205	2.309	2.257	2.118	2.014			
	Ebn at Noon	201	234	244	259	266	264	255	251	252	227	185	170			
	Edh at Noon	24	35	46	43	43	41	43	40	32	27	23	20			
(44) All-Sky Solar Radiation	RadAvg	358	696	1156	1472	1861	1975	1789	1521	1108	590	284	230			
	RadStd	28	30	97	123	159	105	135	108	94	75	35	33			

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	+1.08	+1.51	N/A	N/A	N/A	N/A	(46)
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page